

Simple Linear Regression

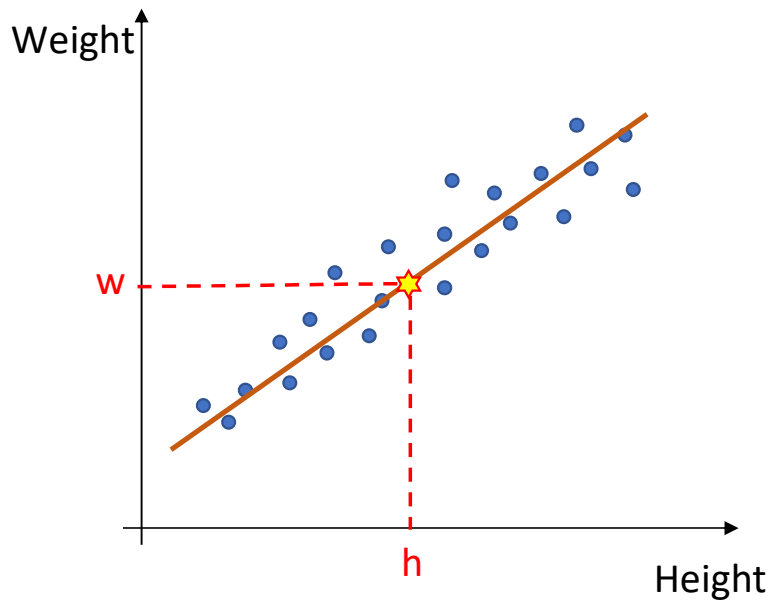
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OUTLINE

- Simple Linear Regression: Intuition
- Hypothesis function for Simple Linear Regression
- Mean Square Error Loss / Cost function
- Intuition of Cost Function
- Ordinary Least Square Regression

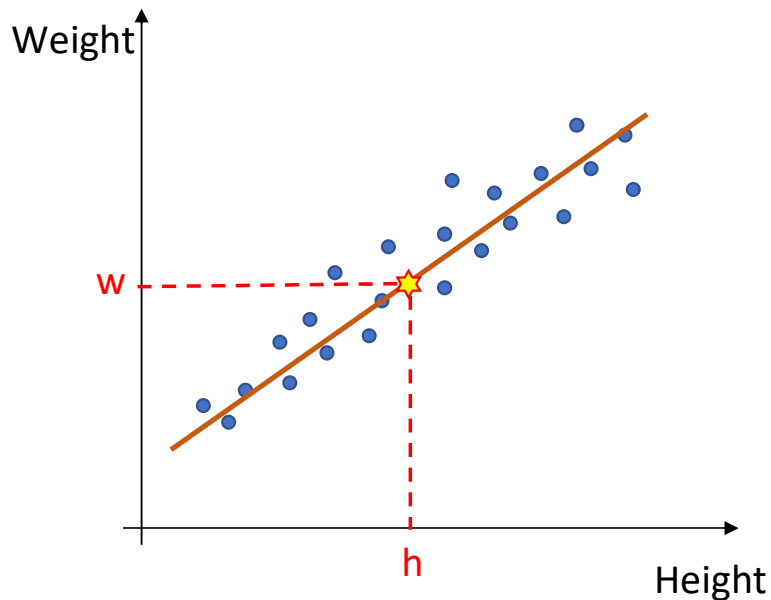
Simple Linear Regression : Intuition



- Consider the scatter plot of the Weight vs. Height of adults as shown beside.
- The trend or the form of the relationship is strongly positive.
- Now suppose we wish to estimate the **weight** of a person just by knowing his/ her **height**.
- In order to do so we first fit a straight line through our data points.

- Then from the graph, knowing the height we can find the weight of the corresponding person.
- Hence, we are intending to find out the **equation of the straight line** that *best* describes the relationship between Weight and Height.

Simple Linear Regression : Intuition



- There is only one predictor/input variable (Height) and one target variable (Weight) and we are intending to find out a relationship of the form:

$$y = \theta_0 + \theta_1 x$$

Here, y is the target variable and x is the predictor variable

- We have to find out θ_0 and θ_1 , such that the straight line $y = \theta_0 + \theta_1 x$ fits into our dataset **best**.
- This is called Simple Linear Regression, because it has only one predictor variable and the relationship among target and predictor variable is linear.

Simple Linear Regression : Hypothesis

Simple Linear Regression Model with Single Predictor

$$\underbrace{y}_{\text{Target Variable}} = \underbrace{\theta_0}_{\text{Intercept}} + \underbrace{\theta_1}_{\text{Slope}} \underbrace{x}_{\text{Predictor Variable}} + \underbrace{\varepsilon}_{\text{Random Residual Error}}$$

θ_0 and θ_1 are called model parameters

The diagram shows the equation $y = \theta_0 + \theta_1 x + \varepsilon$. Brackets are placed under each term: y , θ_0 , θ_1 , x , and ε . Lines connect these brackets to five boxes below: 'Target Variable' for y , 'Intercept' for θ_0 , 'Slope' for θ_1 , 'Predictor Variable' for x , and 'Random Residual Error' for ε . To the right of the equation, a text label states that θ_0 and θ_1 are called model parameters.

- We use our sample data to find estimates for the coefficients/ model parameters θ_0 and θ_1 i.e.: $\widehat{\theta}_0$ and $\widehat{\theta}_1$.
- We can then **predict** what the value of y should be corresponding to a particular value for x by using the Least Squares Prediction Equation (also known as our **hypothesis function**):

$$\widehat{y} = \widehat{\theta}_0 + \widehat{\theta}_1 x \quad \text{Where } \widehat{y} \text{ is our prediction for } y$$

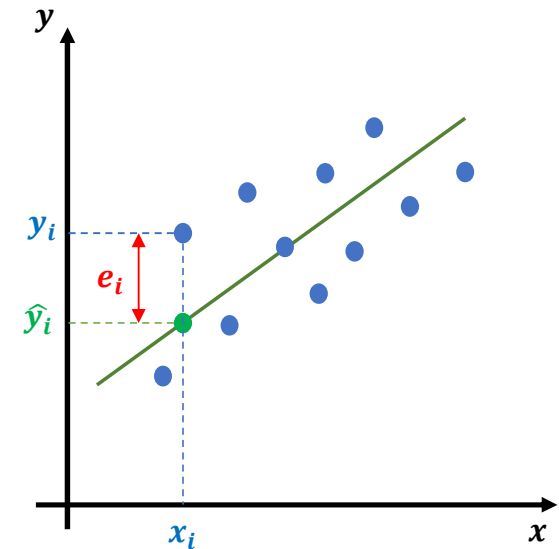
Simple Linear Regression : Cost Function

Residuals and Residual Sum of Squares:

- For i^{th} sample $\langle x_i, y_i \rangle$ the predicted value of y_i is $\hat{y}_i$, Which we obtain from the equation $\hat{y}_i = \hat{\theta}_0 + \hat{\theta}_1 x_i$
- Then, $e_i = y_i - \hat{y}_i$ (actual – predicted) represents the i^{th} residual.
- We define **Residual Sum of Squares (RSS)** as:

$$RSS = \sum_{i=1}^m e_i^2 = \sum_{i=1}^m (y_i - \hat{y}_i)^2 = \sum_{i=1}^m (y_i - (\hat{\theta}_0 + \hat{\theta}_1 x_i))^2$$

There are total m no. of samples



Simple Linear Regression : Cost Function

Mean Square Error Cost Function:

- We can define the cost function as:

$$J(\widehat{\theta}_0, \widehat{\theta}_1) = \frac{1}{2} \frac{RSS}{\text{Number of training samples}} = \frac{1}{2m} \sum_{i=1}^m (y_i - (\widehat{\theta}_0 + \widehat{\theta}_1 x_i))^2$$

Here a factor $\frac{1}{2}$ is multiplied just for computational simplicity. Otherwise, the cost function $J(\widehat{\theta}_0, \widehat{\theta}_1)$ is nothing but **mean or average of the Residual sum of squares**. (also known as **Mean Square Error (MSE)**).

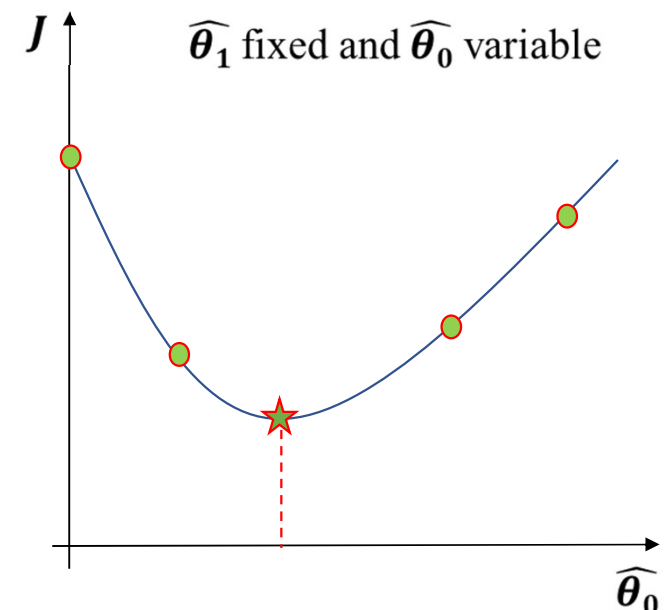
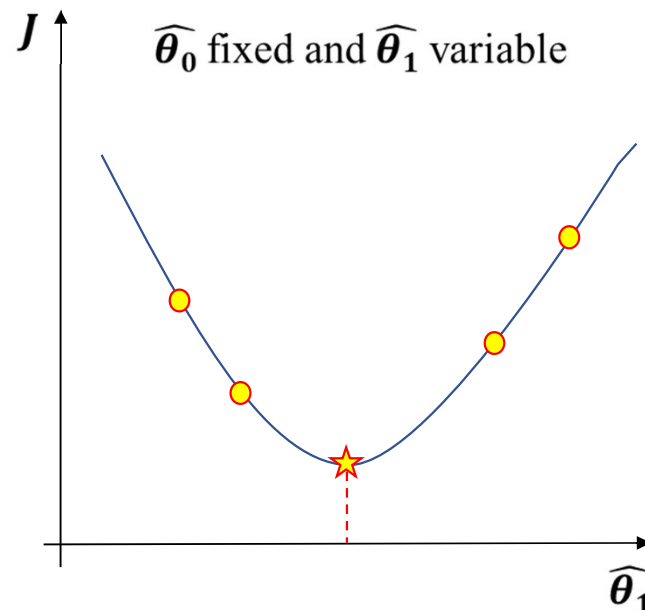
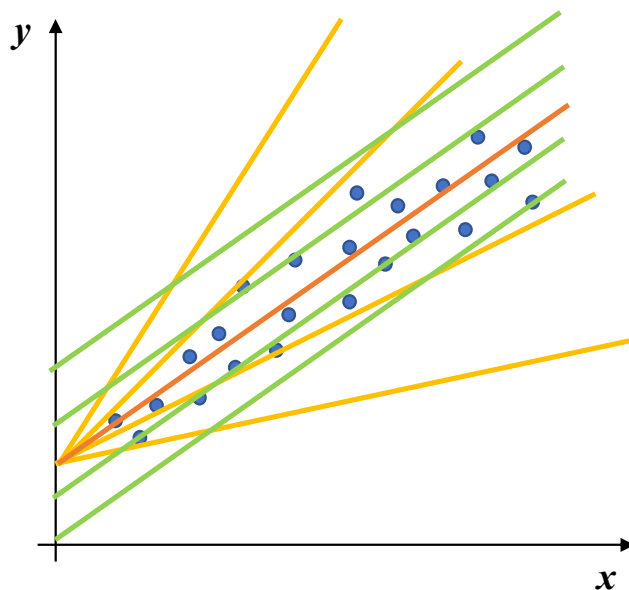
Our Objective:

- To find the suitable values of $\widehat{\theta}_0$ and $\widehat{\theta}_1$ such that the cost function $J(\widehat{\theta}_0, \widehat{\theta}_1)$ is minimized, in other words the Residual Sum of Square (**RSS**) is minimized. Then the straight line $\widehat{y} = \widehat{\theta}_0 + \widehat{\theta}_1 x$ will fit our data **best**. This is called **least squares fit**.

Simple Linear Regression : Cost Function

Intuition of Cost Function:

Consider the example of single predictor variable where the hypothesis function is $\hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x$ and the cost function is $J(\hat{\theta}_0, \hat{\theta}_1) = \frac{1}{2m} \sum_{i=1}^m (y_i - (\hat{\theta}_0 + \hat{\theta}_1 x_i))^2$. Now we keep one parameter fixed and vary other. Let's see how $J(\hat{\theta}_0, \hat{\theta}_1)$ varies.



- Our objective is to find the values of the parameters for which the cost function is minimized.

Simple Linear Regression : OLS fit

Solving for the best fit: Ordinary Least Squares (OLS) Regression:

- We have to Minimize **RSS** or $J(\widehat{\theta}_0, \widehat{\theta}_1)$ with respect to $\widehat{\theta}_0$ and $\widehat{\theta}_1$
- Hence we have to do, $\frac{\partial}{\partial \widehat{\theta}_0} (RSS) = 0$ and $\frac{\partial}{\partial \widehat{\theta}_1} (RSS) = 0$
- By solving the above two equations we get the following value of $\widehat{\theta}_1$ and $\widehat{\theta}_0$:

$$\widehat{\theta}_1 = \frac{\sum_{i=1}^m (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^m (x_i - \bar{x})^2} = r_{xy} \frac{\sigma_y}{\sigma_x} \quad \text{and} \quad \widehat{\theta}_0 = \bar{y} - \widehat{\theta}_1 \bar{x}$$

where, $\bar{x}$ is the mean of predictor variable x and $\bar{y}$ is the mean of target variable y

σ_x is the standard deviation of x and σ_y is the standard deviation of y

*and r_{xy} is the **correlation coefficient** between x and y .*

Ordinary Least Square : Proof

Cost Function: $J(\hat{\theta}_0, \hat{\theta}_1) = \frac{1}{2m} \sum_{i=1}^m (y_i - (\hat{\theta}_0 + \hat{\theta}_1 x_i))^2 = \frac{1}{2m} \sum_{i=1}^m ((\hat{\theta}_0 + \hat{\theta}_1 x_i) - y_i)^2$

To minimize the cost function, we have to set: $\frac{\partial J}{\partial \hat{\theta}_0} = 0$ and $\frac{\partial J}{\partial \hat{\theta}_1} = 0$

$$\frac{\partial J}{\partial \hat{\theta}_0} = 0$$

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^m \frac{\partial}{\partial \hat{\theta}_0} ((\hat{\theta}_0 + \hat{\theta}_1 x_i) - y_i)^2 = 0$$

$$\Rightarrow \frac{2}{2m} \sum_{i=1}^m ((\hat{\theta}_0 + \hat{\theta}_1 x_i) - y_i) = 0$$

$$\Rightarrow \frac{1}{m} \sum_{i=1}^m \hat{\theta}_0 + \hat{\theta}_1 \frac{1}{m} \sum_{i=1}^m x_i - \frac{1}{m} \sum_{i=1}^m y_i = 0$$

$m\hat{\theta}_0 \qquad \bar{x} \qquad \bar{y}$

$$\Rightarrow \hat{\theta}_0 + \hat{\theta}_1 \bar{x} - \bar{y} = 0 \Rightarrow \hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x}$$

$$\frac{\partial J}{\partial \hat{\theta}_1} = 0$$

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^m \frac{\partial}{\partial \hat{\theta}_1} ((\hat{\theta}_0 + \hat{\theta}_1 x_i) - y_i)^2 = 0$$

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^m \frac{\partial}{\partial \hat{\theta}_1} ((\bar{y} - \hat{\theta}_1 \bar{x} + \hat{\theta}_1 x_i) - y_i)^2 = 0$$

$$\Rightarrow \frac{1}{2m} \sum_{i=1}^m \frac{\partial}{\partial \hat{\theta}_1} (\hat{\theta}_1 (x_i - \bar{x}) - (y_i - \bar{y}))^2 = 0$$

$$\Rightarrow \frac{2}{2m} \sum_{i=1}^m (\hat{\theta}_1 (x_i - \bar{x}) - (y_i - \bar{y})) (x_i - \bar{x}) = 0$$

Ordinary Least Square : Proof

$$\Rightarrow \frac{2}{2m} \sum_{i=1}^m \left(\hat{\theta}_1 (x_i - \bar{x}) - (y_i - \bar{y}) \right) (x_i - \bar{x}) = 0$$

$$\Rightarrow \hat{\theta}_1 \sum_{i=1}^m (x_i - \bar{x})^2 = \sum_{i=1}^m (x_i - \bar{x}) (y_i - \bar{y})$$

$$\Rightarrow \hat{\theta}_1 = \frac{\sum_{i=1}^m (x_i - \bar{x}) (y_i - \bar{y})}{\sum_{i=1}^m (x_i - \bar{x})^2}$$

$$\Rightarrow \hat{\theta}_1 = \frac{\frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x}) (y_i - \bar{y})}{\frac{1}{m-1} \sum_{i=1}^m (x_i - \bar{x})^2} \Rightarrow \frac{Cov(X, Y)}{Var(X)}$$

$$\Rightarrow \hat{\theta}_1 = \frac{r_{xy} \sigma_x \sigma_y}{\sigma_x^2}$$

$$\Rightarrow \hat{\theta}_1 = r_{xy} \frac{\sigma_y}{\sigma_x}$$

$$\text{Correlation Coefficient} = r_{xy} = \frac{Cov(X, Y)}{\sigma_x \sigma_y}$$

$$\Rightarrow Cov(X, Y) = r_{xy} \sigma_x \sigma_y$$

$$\text{also, } Var(X) = \sigma_x^2$$

Where, σ_x = std. deviation of X and
 σ_y = std. deviation of Y

Thank You